Student Performance Trend Analysis (Case Study 17)

Problem Statement

Compare a student's marks in multiple tests and report the change between each pair of tests.

Objectives

Use lists, loops, indexing, and if/else conditions.

Assumptions

Marks are entered in the same subject/test and are between 0 and 100.

Algorithm

1. Read marks for three tests into a list. 2. Compare each mark with the previous mark. 3. Print improved, declined, or
unchanged. 4. Print the average.

Flowchart (Text Form)

START → Read marks list → Compare adjacent marks → Print trend → Calculate average → STOP.

Pseudocode

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```

Python Source Code

```
marks = []
for i in range(3):
    mark = float(input("Enter test mark (0-100): "))
    marks.append(mark)
for i in range(1, len(marks)):
    change = marks[i] - marks[i - 1]
    if change &gt; 0:
        print("Test", i + 1, ": improved by", change)
    elif change &lt; 0:
        print("Test", i + 1, ": declined by", abs(change))
    else:
        print("Test", i + 1, ": unchanged")
print("Average mark:", round(sum(marks) / len(marks), 2))
```

Test Case and Expected Output

Input marks: 60, 75, 70
Expected: Test 2 improved by 15; Test 3 declined by 5; Average mark: 68.33

Conclusion

The program shows whether performance is improving and calculates the average mark.